

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer

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Professional Summary

Motivated data professional with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data product solutions for complex challenges. Expert in managing the software development lifecycle, emphasizing stakeholder engagement, KPI definition, team building, and scalable deployment. Ability to articulate complex data insights to diverse stakeholders, ensuring clear understanding and informed decision-making.

Education

Bachelor of Science in Mechanical Engineering – Arizona State University — Tempe, AZ — 06/2003 to 05/2006

Programming Languages

Python, R, SQL, Snowflake SQL, Transact SQL, BASH, Powershell, MATLAB

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Architectural Design, Artificial Intelligence, AWS, CI/CD, Computer Science, Continuous Integration, DAG, Database Administrator, Data Gathering, DBA, Deep Learning, Design of Experiments, DevOps, Developer Tools, Distributed Computing, Docker, ECR, Embedding, ETL, Advanced Excel, Financial Forecasting, Generative AI, Git, GitHub Actions, GitHub Copilot, LangChain Expression Language, LCEL, Linear Algebra, Large Language Models, LLMs, Mathematics, Multivariate Calculus, Pandas, Power BI, PostgreSQL, Problem Solving, PyTorch, RAG, Redshift, Research, Scikit-learn, S3, Sagemaker, Scripting Language, Scrum, SDLC, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Step Functions, Streamlit, Tableau, TensorFlow, Transformers, Version Control

Soft Skills

Adaptability, Business Requirements, Coaching, Communication Skills, Creative, Growth Mindset, Leadership, Mentoring, Multicultural Collaboration, Strategic Thinking, Thought Leadership, Willing to Learn

Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ — 09/2022 to Present

- Architect and developer of end-to-end data science initiatives, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Performed data mining on structured, semi-structured, and unstructured data to assemble a training dataset from multiple sources, considering thousands of parameters derived from dealer and product information
- Used an LLM to find similarities in service jobs and feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Utilized Python and Autogluon to train the AI model, leveraging machine learning algorithms such as neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and led to more accurate predictions of equipment repair times
- Engineered and deployed robust MLOps pipelines and cloud computing using AWS and Snowflake, as a database administrator, and established best practices for cloud and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact
- Implemented robust information management strategies to ensure data integrity and accessibility for the Snowflake data lake, improving data reliability and supporting informed decision-making

Reliability Analytics Manager — Amazon.com — Tempe, AZ — 02/2022 to 09/2022

- Applied advanced analytics to create a new KPI for autonomous robots through big data analysis and wrote an influential white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Utilized advanced data science techniques to extract actionable insights, driving strategic decision-making and improving operational outcomes
- Designed and optimized regression and recommendation systems, achieving two-time champion status in Amazon Machine Learning University competitions, which improved predictive accuracy and recommendation quality in practical applications
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager — Empire-Cat — Mesa, AZ — 01/2012 to 02/2022

- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days
- Developed and deployed an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting
- Directed a team of up to 8 engineers and served as a client-facing expert for mining equipment, fostering a data-driven approach to solve equipment performance issues, which improved fleet efficiency and client satisfaction
- Applied the DMAIC Six Sigma methodology with complex problem-solving using R and SQL to identify opportunities for improving operation, manufacturing, and maintenance of a large off-highway truck fleet, yielding \$1.2 billion in savings over six years as Reliability Engineering Manager prior to promotion
- Business acumen to build trust and enhance stakeholder relationships to promote data-based decisions. Partnered on design of experiments to improve reliability, safety and durability.

Certifications

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **CSCMP:** Supply Chain Foundations Demand Planning Professional Certificate
- **Data Camp:** AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate
- **Empire-Cat:** Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IUOE Local 150:** Diesel Mechanic, Bulldozers, Class B CDL, Cranes, Forklifts, Heavy Equipment Operator, MSHA, OSHA, Scrapers, Skid-steers, Telehandlers, Wheel Loaders Shielded Metal Arc Welding (SMAW)
- **Komatsu North America:** Certified Technical Communicator
- **LinkedIn Learning:** Data Engineering with dbt
- **Society of Automotive Engineers:** Cost and Finance for Engineers
- **Udemy:** Streamlit for Snowflake Masterclass